2-Bedroom Boys' Quarters

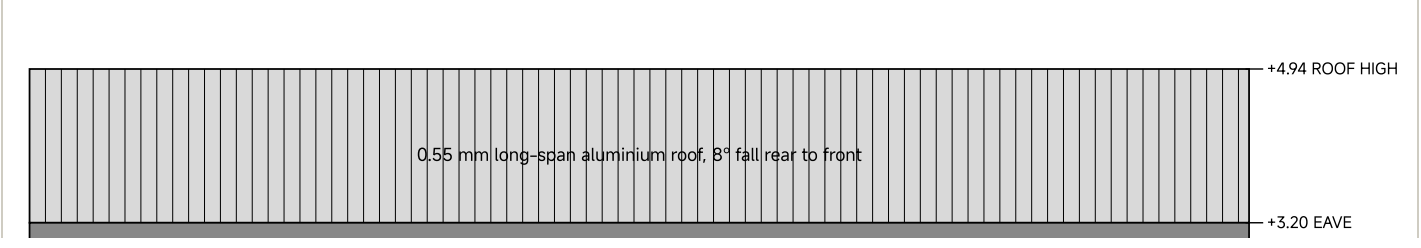
Preliminary design drawings, Nigeria. Ground floor only. Dimensions in metres unless noted. Proposed values need professional approval. Not for construction.

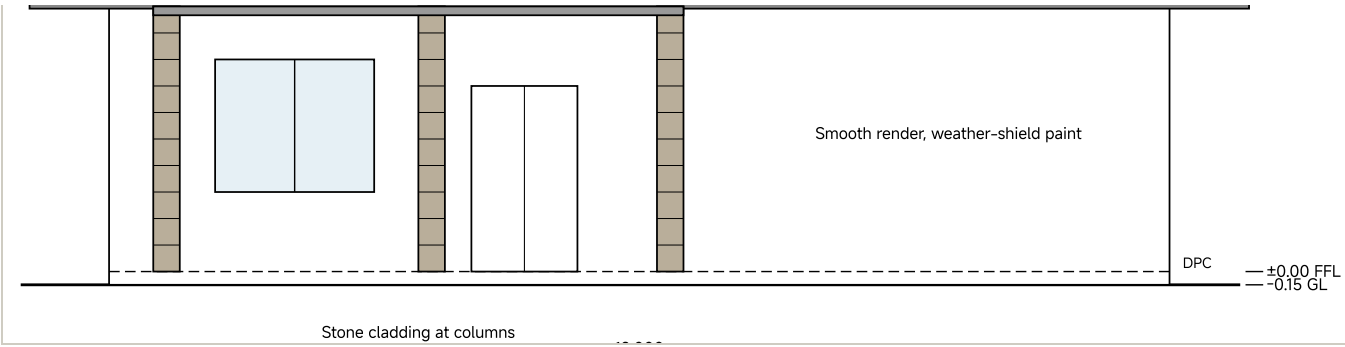
A-001 Floor plan



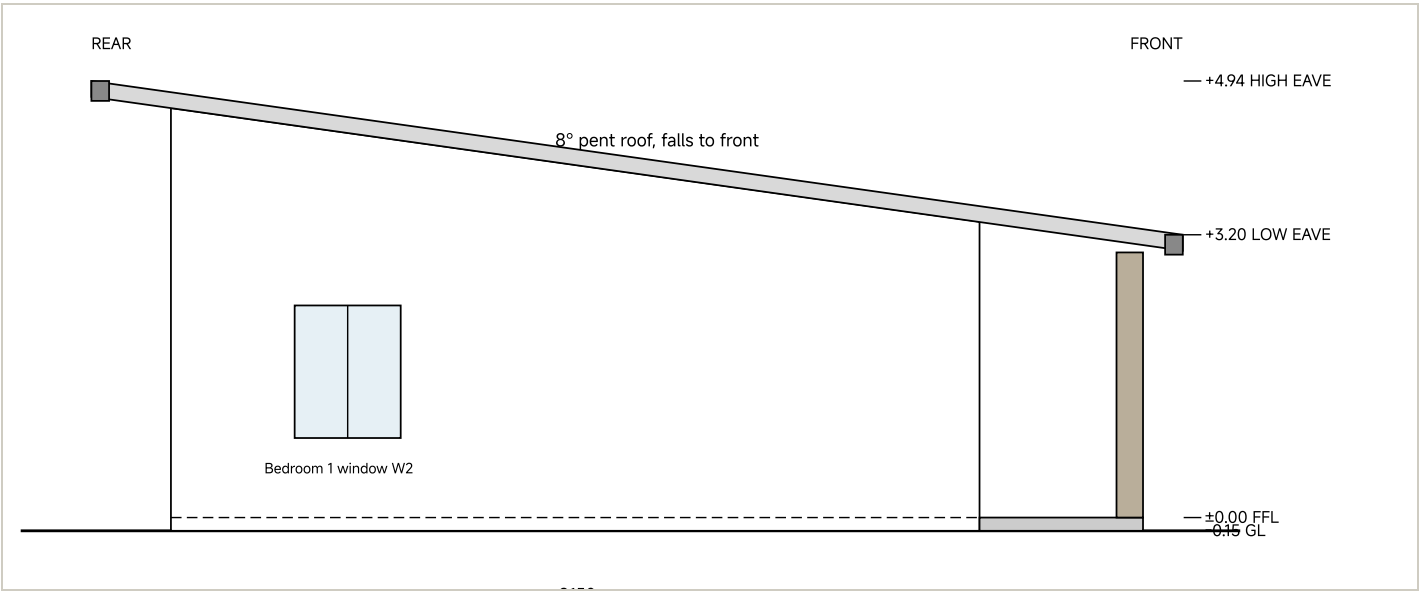
A-001 GROUND FLOOR PLAN · 1:100 (schematic, wall thickness 225 ext / 150 int)

A-003 Front elevation

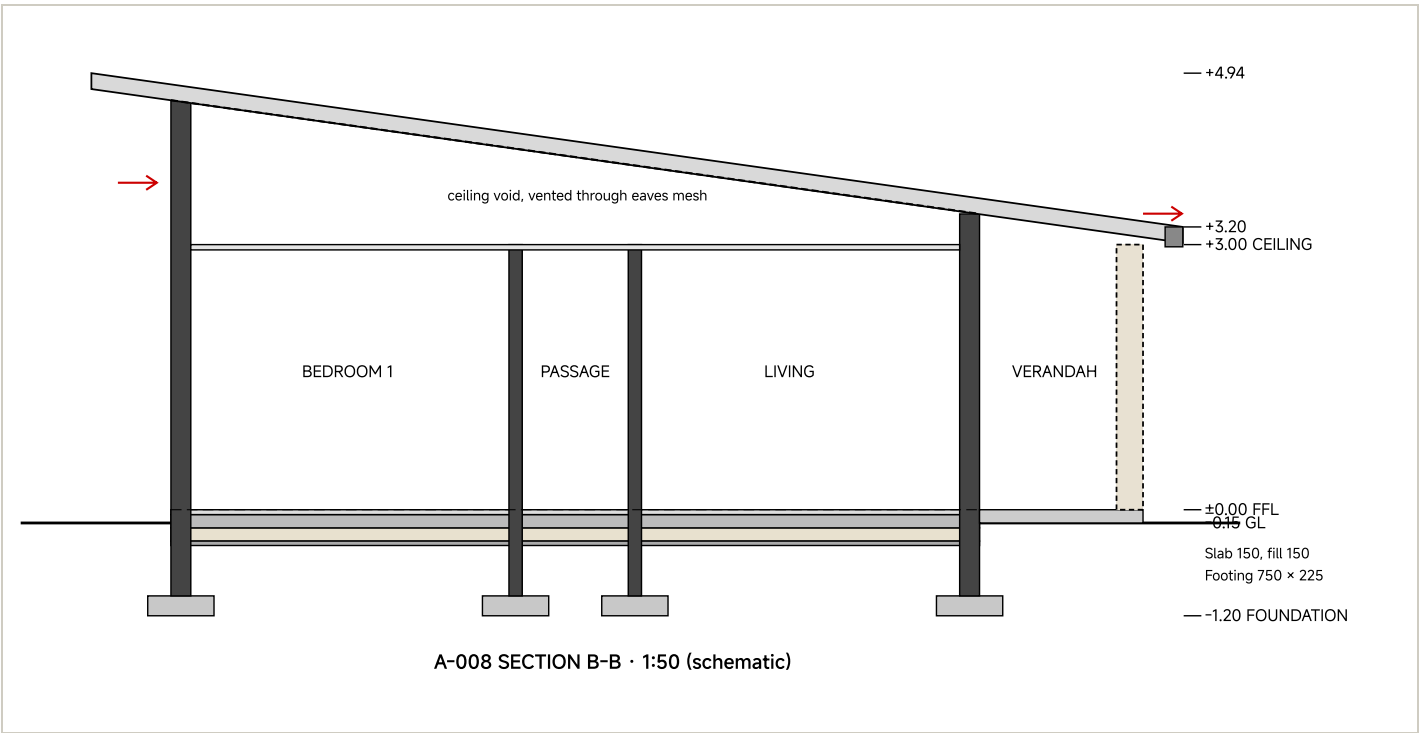




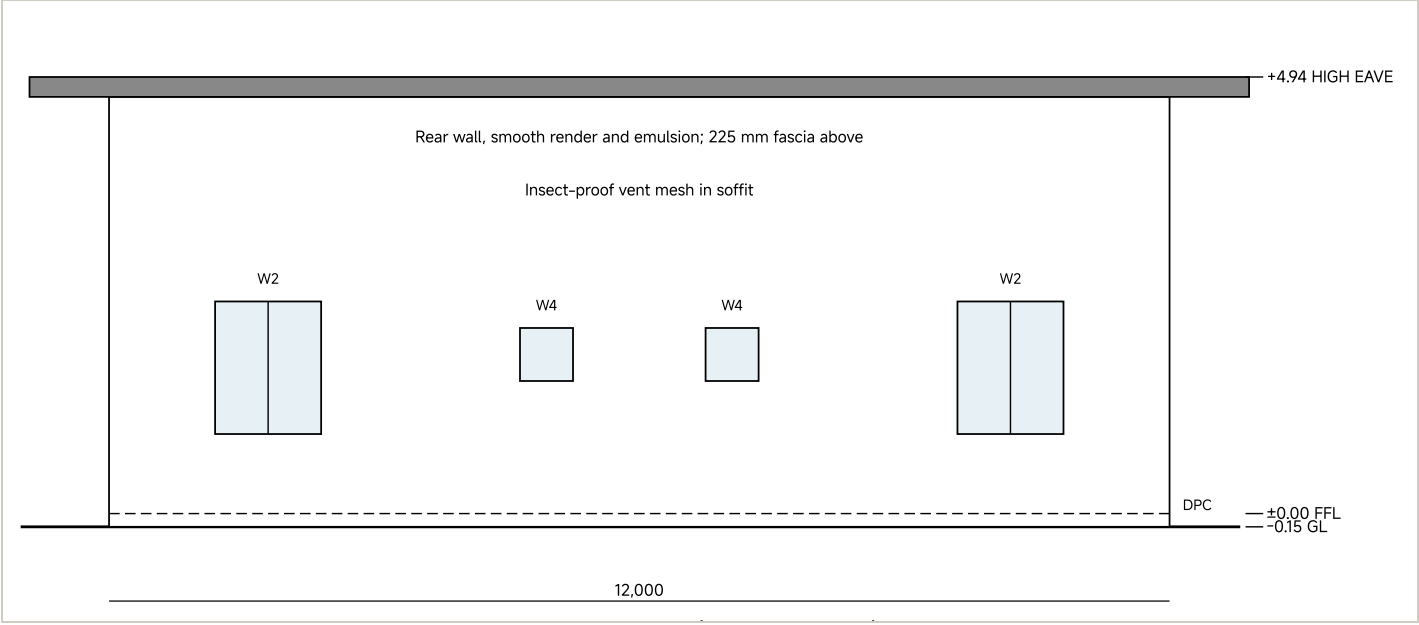
A-005 Left elevation



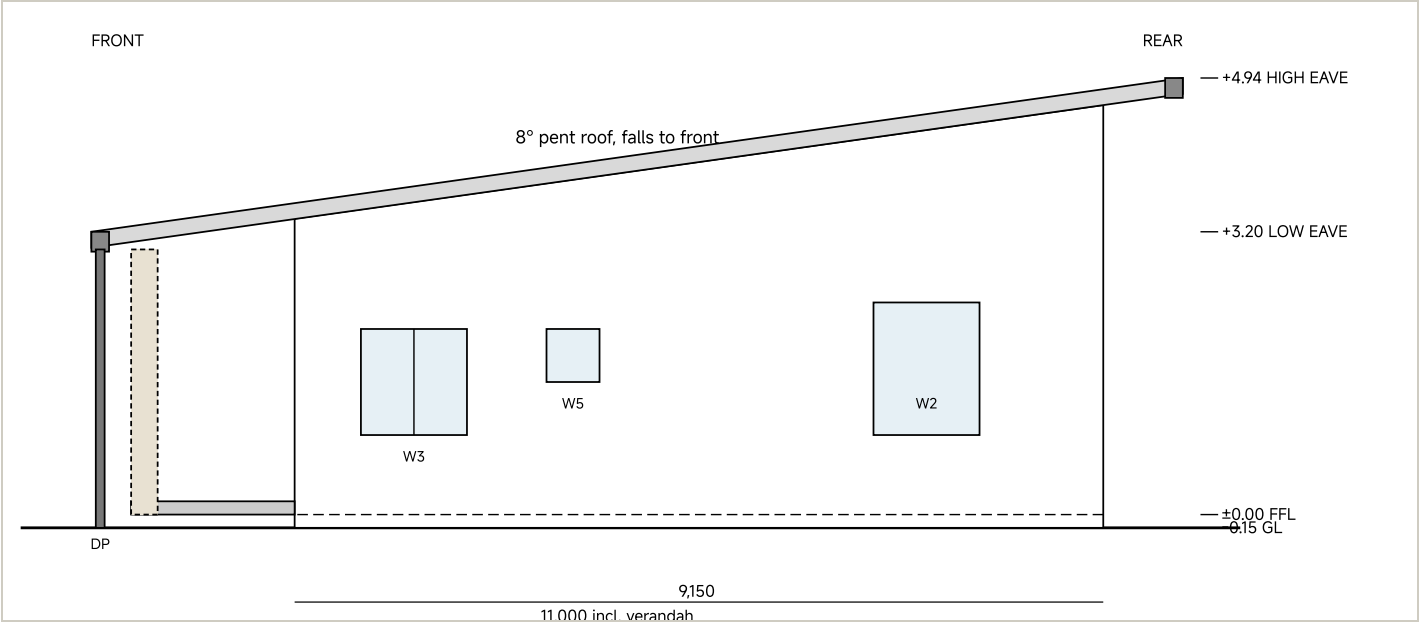
A-008 Section B-B (cross-section at x = 1.00 m)



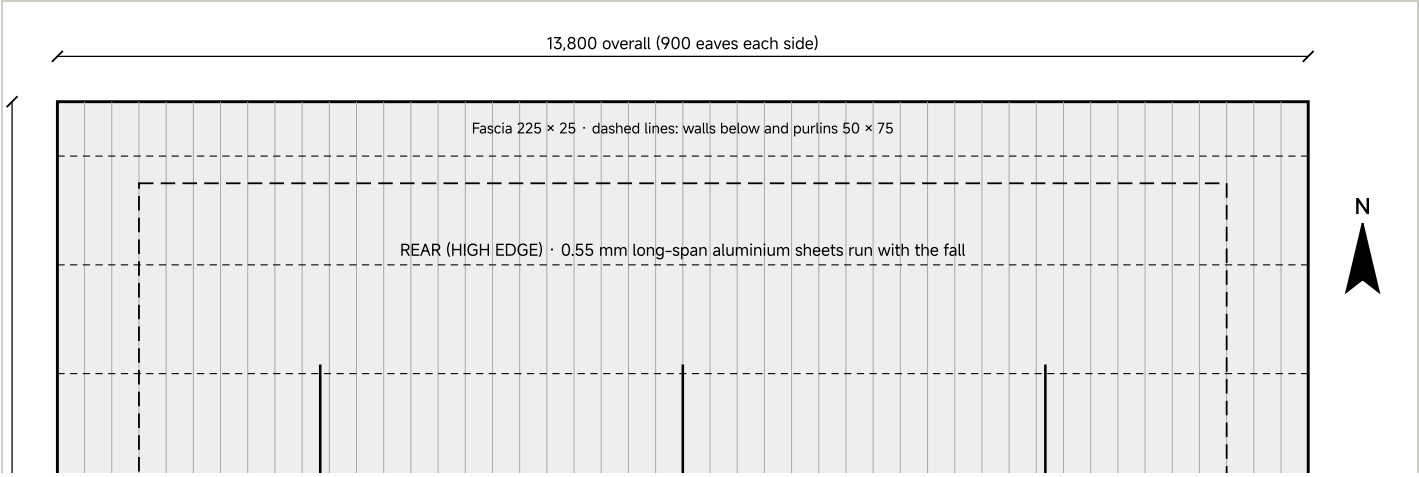
A-004 Rear elevation

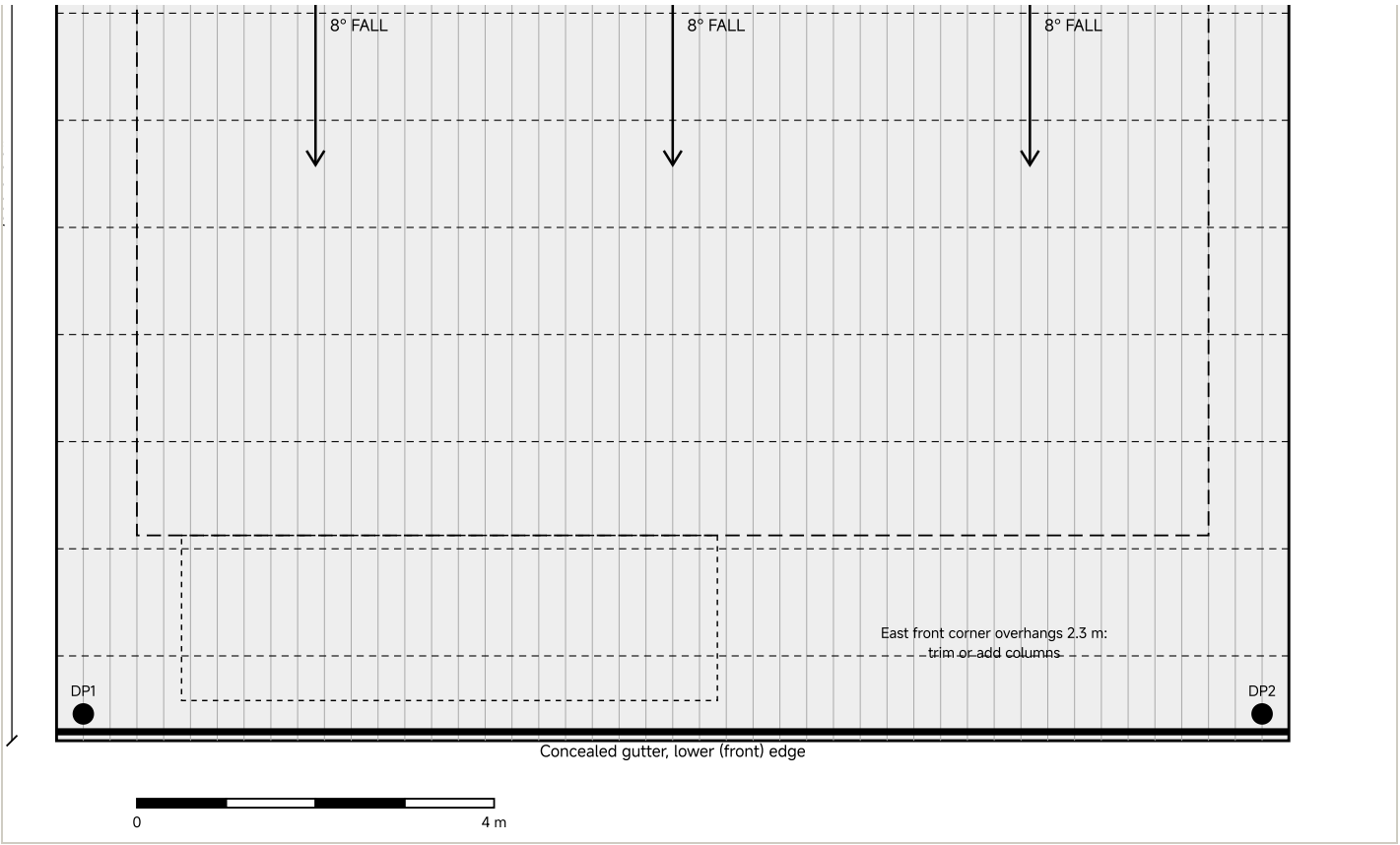


A-006 Right elevation

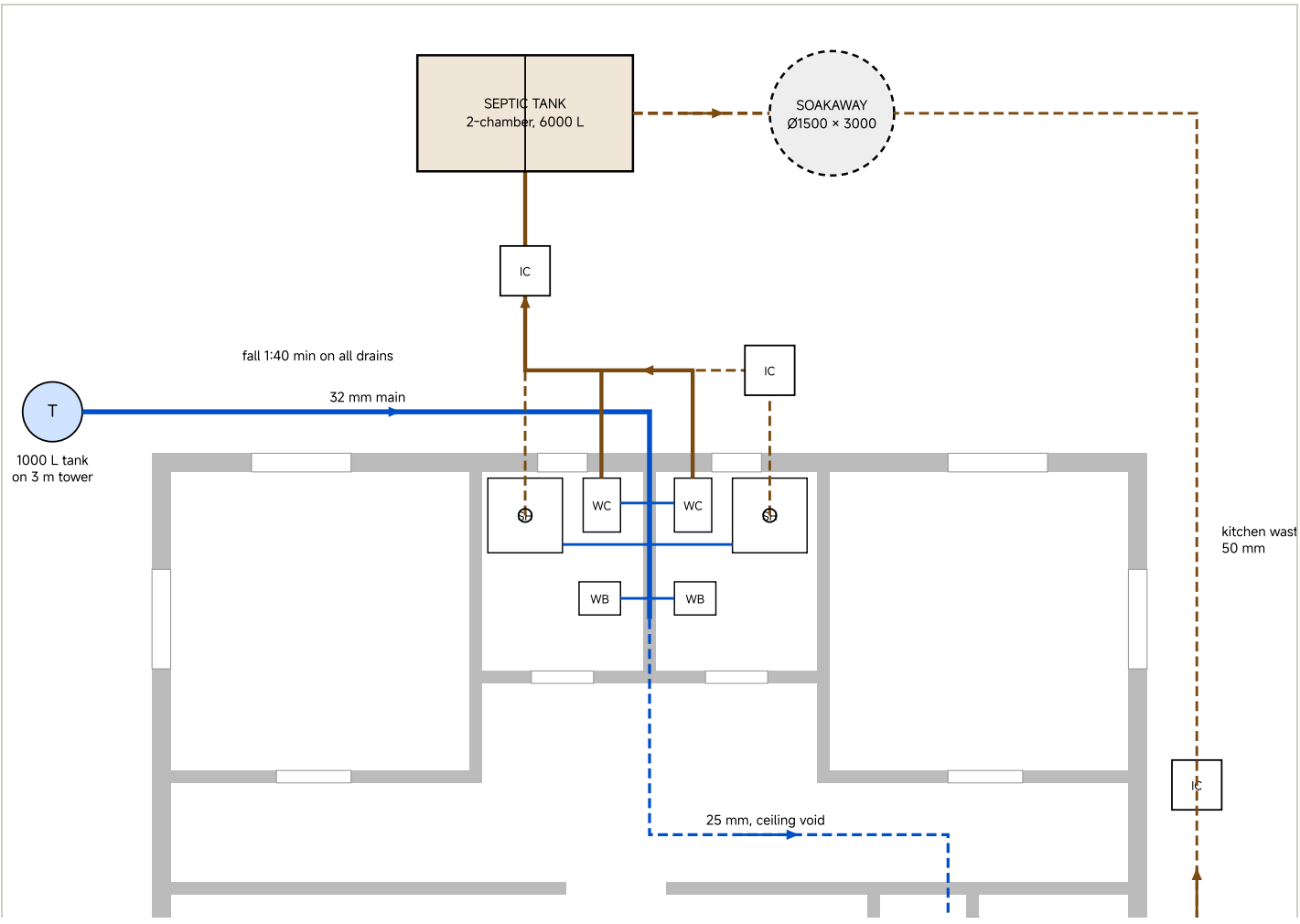


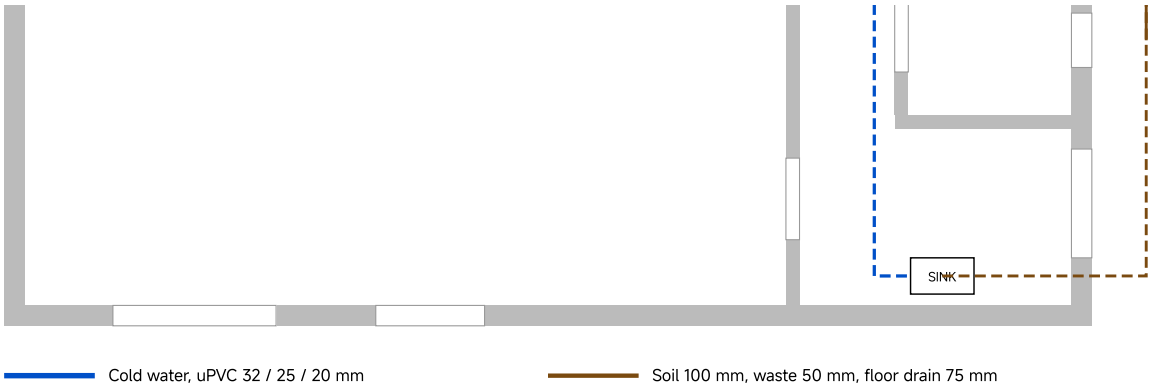
A-002 Roof plan





P-001 Plumbing layout

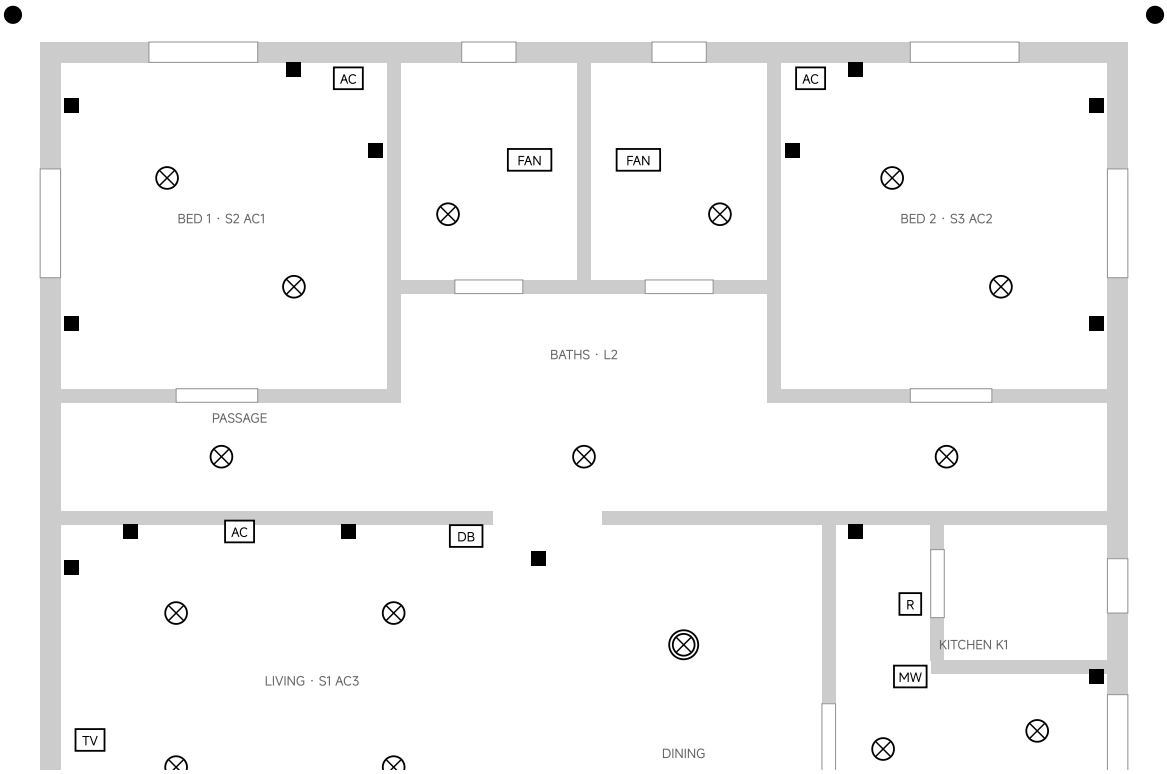


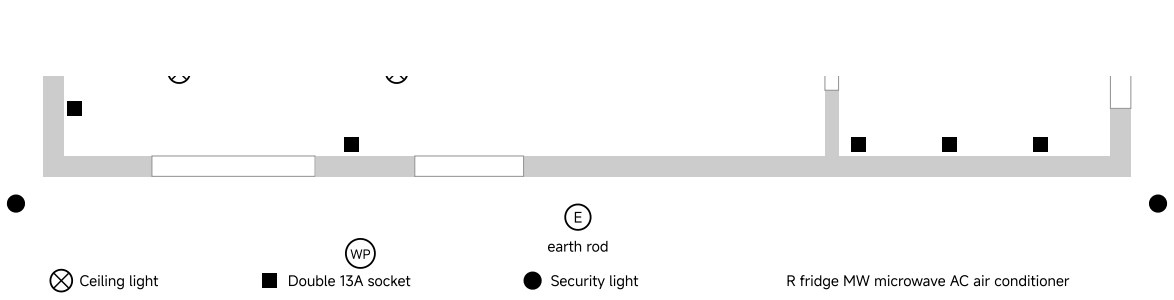


P-001 PLUMBING LAYOUT · 1:100 (schematic)

Item	Basis
Storage	1000 L tank on a 3 m tower, west of the building (proposed position)
Supply pipes	Main 32 mm, branches 25 mm, fixture supply 20 mm, uPVC. Kitchen branch runs in the ceiling void.
Drainage	Soil 100 mm, waste 50 mm, floor waste 75 mm, uPVC. Fall 1:40 minimum.
Disposal	Two-chamber septic tank, 6000 L minimum, then soakaway 1500 mm diameter, 3000 mm deep. Kitchen waste also goes to the soakaway.

E-001 Electrical layout



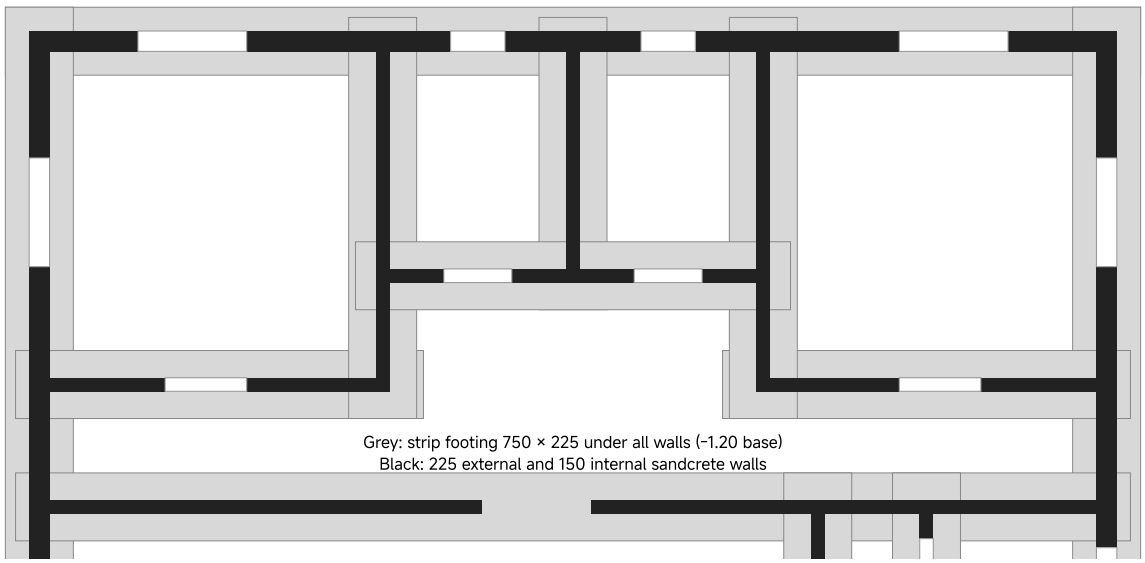


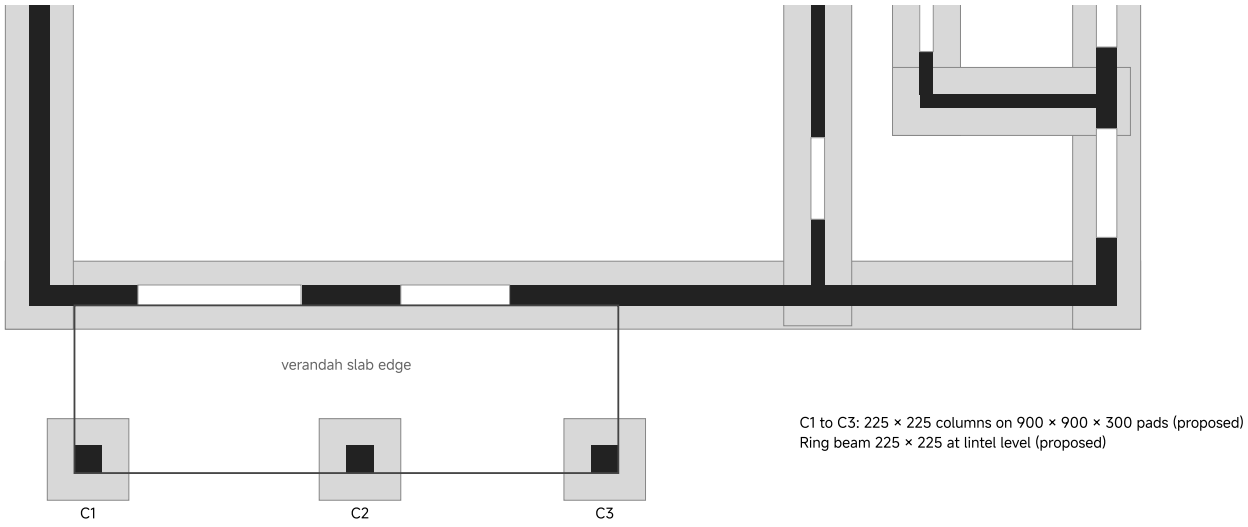
E-001 ELECTRICAL LAYOUT · 1:100 (schematic)

Circuit	Serves	Cable
L1	Living (4), dining pendant, verandah, 4 security lights	1.5 mm²
L2	Bedrooms (2 each), kitchen (2), bathrooms (1 each) with fans, passage (3, proposed)	1.5 mm²
S1	Living and dining, 6 double sockets, TV point	2.5 mm²
S2, S3	Bedroom 1 and Bedroom 2, 4 double sockets each	2.5 mm²
K1	Kitchen, 5 sockets	2.5 mm²
D1	Refrigerator and microwave points	2.5 mm²
AC1 to AC3	Bedroom 1, Bedroom 2, living room	4 mm²
Earthing	16 mm² cable to a 16 × 2400 mm copper-bonded rod	16 mm²

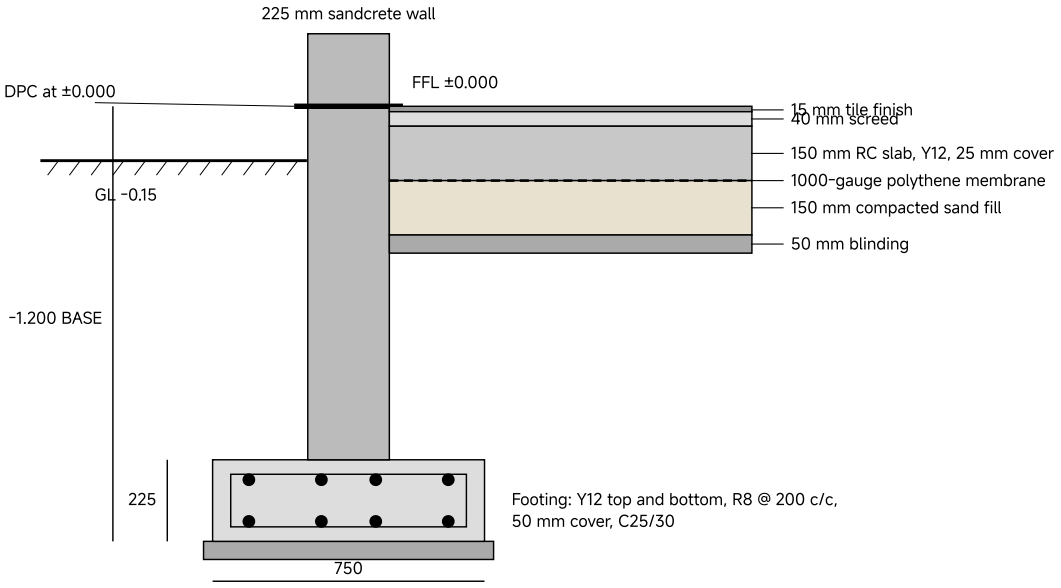
Single phase 230 V. Main distribution board at the entrance lobby. Breaker ratings and load calculation are for the electrical engineer.

S-001 Structural layout and details





S-001 FOUNDATION AND WALL LAYOUT · 1:100 (schematic)



S-001 TYPICAL FOOTING AND FLOOR DETAIL · schematic, bars shown diagrammatically

Item	From your spec
Concrete	C25/30. Foundation 1:3:6 listed as well, to be confirmed.
Strip footing	750 × 225, Y12 top and bottom, R8 @ 200 c/c, 50 mm cover, 1200 mm minimum depth, 50 mm blinding
Floor slab	150 mm reinforced, Y12 bars, 25 mm cover, on 150 mm sand fill and polythene membrane
Walls and DPC	225 mm external and 150 mm internal sandcrete blocks, bituminous DPC, anti-termite treatment before the slab
Proposed, not in spec	225 × 225 ring beam and lintels, three 225 × 225 verandah columns on 900 × 900 pads. Reinforcement and sizes to be set by the engineer.

Room schedule (internal, proposed)

Space	Size (m)	Area (m²)
Bedroom 1	3.60 × 3.60	12.96
Bedroom 2	3.60 × 3.60	12.96
Living room	5.40 × 3.60	19.44
Dining (open to living)	3.00 × 3.60	10.80
Kitchen (includes pantry)	3.00 × 3.60	10.80
Pantry	1.80 × 1.50	2.70
Bathroom 1 and 2 (each)	1.95 × 2.40	4.68
Utility / store bay	4.05 × 1.05	4.25
Passage	1.20 × 11.55	13.86
Verandah	6.00 × 1.85	11.10
Internal area, enclosed	12.00 × 9.15 external	**≈ 94**

Door schedule

Tag	Use	Size (mm)	Qty
D1	Main entrance, swings out	1200 × 2100	1
D2	Bedrooms, swing in	900 × 2100	2
D3	Kitchen, from dining	900 × 2100	1
D4	Bathrooms, swing in	750 × 2100	2
D5	Pantry	750 × 2100	1

Window schedule (sill 900 mm)

Tag	Type and location	Size (mm)	Qty
W1	Living, sliding, front wall	1800 × 1500	1
W2	Bedroom casement, side and rear walls	1200 × 1500	4
W3	Kitchen casement, east wall	1200 × 1200	1
W4	Bathroom louvre, rear wall	600 × 600	2
W5	Pantry louvre, east wall	600 × 600	1

Changes and assumptions

- **Footprint:** 12.00 × 11.00 m including the verandah (main block 12.00 × 9.15). An 11.5 × 9 m footprint could not hold the 1.2 m passage, verandah and all rooms.
- **Roof:** 8° pent roof, high at the rear. Roof run is 12.35 m, so the rise is about 1.74 m. Highest point is +4.94, replacing the +4.30 in the spec.
- **Store:** provided as an open utility/store bay. D5 now serves the pantry.
- **Kitchen window W3** moved from the rear wall to the east wall, because the rear wall of the kitchen is the passage.
- **DPC:** drawn at ±0.000 (150 mm above ground). The spec's level table shows +0.150, which conflicts. Please confirm.
- **Orientation:** assumed front faces south, with north up. Please confirm.
- **Concrete:** spec lists both 1:3:6 and C25/30 for foundations. The structural engineer should confirm.
- **Roof edge:** the roof is one rectangle, so the front eave overhangs the wall by 2.3 m on the east half, beyond the 6 m verandah. It needs extra columns or a cut-back before it can be built. Downpipe positions are also provisional.
- **Louvres W4 and W5** are drawn high-level, with sill at 1.50 m, for privacy.
- **Services:** tank, septic tank and soakaway positions are provisional. Clearances to the boundary and building, soil type and water table must be checked on site. Supply and drain pipes cross near the rear wall at different levels.
- Structural, electrical and plumbing design and all approvals must be done by registered professionals.